



DATA & AI CHALLENGE · INTELLIGENT CANDIDATE DISCOVERY & RANKING

Reading the profile, not the keywords.

A hybrid, CPU-only ranker that beats keyword-stuffers,
surfaces plain-language talent, and avoids the honeypots.

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THE PROBLEM

Keyword filters can't see what matters

- **Recruiters miss the right person — keyword filters are blind to real fit.**
The JD is written to defeat keyword matching; the right answer is reasoning about what it means.
- **The dataset is adversarial by design.**
Keyword stuffers, plain-language strong candidates, behavioral twins, and ~80 'impossible' honeypots.
- **The shipped sample submission deliberately falls for it.**
It ranks an HR Manager #1 and real ML Engineers at #27 / #48 / #99 — purely on AI-skill counts.
- **Goal: a shortlist a recruiter can trust — that survives reproduction + interview.**
NDCG@10 dominates scoring (0.50), so the top-10 must be genuinely excellent.



READING BETWEEN THE LINES

What the role actually needs

FIT — the ideal hire

- 6–8 yrs, applied ML at product companies (not services)
- Shipped search / ranking / retrieval / recsys to real users
- Embeddings + vector/hybrid search + rigorous ranking eval
- Strong NLP/IR; in/near Pune–Noida; actually available

ANTI-FIT — explicit disqualifiers

- Stuffed AI skills but a non-technical real title
- Pure research / academia with no production
- Only recent LangChain-on-OpenAI wrappers
- CV/speech-only · services-only · title-chasers · dormant



ARCHITECTURE

Four interpretable layers, no LLM at inference

$$\text{score} = (0.70 \text{ structured_fit} + 0.30 \text{ semantic_fit}) \times \text{behavioral_modifier} \times \text{honeypot_gate}$$

Structured fit

title coherence
career evidence
skill trust
domain · location

Semantic fit

contrastive:
JD-ideal minus
JD-anti-pattern
static embeds

Behavioral

availability
recency · response
open-to-work
notice period

Honeypot gate

impossible
tenure / skills
removed from
top-100

Two-pass: scalar scoring over 100K, then full facts + grounded reasoning for the top 100. CPU-only, streams the pool.



LAYER 1 — STRUCTURED FIT

The JD's logic, made machine-usable (70% weight)

- **Title coherence — the decisive anti-stuffer signal.**

A 'Marketing Manager' with 9 AI skills scores ~0.06; bullseye AI/search/recsys titles score 1.0.

- **Career evidence — read from descriptions, not skill tags.**

Credits demonstrated retrieval / ranking / production / NDCG-MAP evaluation / NLP-IR work.

- **Skill trust — defeats stuffing.**

Skills reweighted by endorsements x duration x proficiency x platform assessment (raw lists ignored).

- **Plus experience band, product-vs-services, NLP/IR-vs-CV/speech, location, disqualifier penalties.**

Research-only, LangChain-only, and title-chaser flags apply explicit multiplicative penalties.



LAYER 2 — SEMANTIC FIT

Contrastive static embeddings (30% weight, CPU, no GPU)

- **sem = $\cos(\text{candidate}, \text{JD-ideal}) - \cos(\text{candidate}, \text{JD-anti-pattern})$.**

Subtracting similarity to the unwanted profile is what mutes stuffers (close to both anchors).

- **Surfaces plain-language Tier-5s.**

Someone titled 'Software Engineer' who describes building a recommendation system rises on semantics.

- **model2vec static embeddings — distilled for retrieval.**

No transformer forward pass, no torch, no GPU: the latency/quality tradeoff the JD asks about.

- **Pure scikit-learn LSA fallback keeps it reproducible with zero downloads.**



LAYERS 3 & 4 — AVAILABLE & POSSIBLE

Down-weight the unavailable; remove the impossible

Behavioral modifier

- 'Dormant 6 months + 5% response = not available.'
- Bounded multiplier from 23 signals, calibrated to the pool.
- Recency, response rate, open-to-work, notice, demand.

Honeypot gate

- Detects internal impossibilities, not specific IDs.
- Tenure > time elapsed; ≥ 3 expert skills at 0 months.
- 0 honeypots in our top-100 (cap is 10%).



WHAT THE DATA TOLD US

EDA on the full 100K pool drove every threshold

- **Skills are ~uniform random noise.**

Every common skill sits at ~12% of profiles — 'semantic search' as a listed skill means almost nothing.

- **Only ~1,200 of 100,000 have a genuine AI/ML title.**

~68K carry the 12 non-technical 'stuffer' titles; the bullseye titles number in the dozens.

- **Honeypots are rare and detectable.**

We gate 43 internally-impossible profiles pool-wide via tenure/skill contradictions.

- **Availability is a gradient, not a cliff.**

Everyone is ≥ 45 days inactive (median 130) — recency is scaled to the real distribution.



RESULTS

A top-100 a recruiter can trust

95%

based in India

85%

open to work

6.4

mean years exp.

0

honeypots in top-100

Sample top-ranked reasoning (from the candidate's own facts — no hallucination):

- #1 Senior AI Engineer, 5.9 yrs; career history shows building retrieval/search & ranking/recommendation systems in production with NDCG/MAP-style evaluation; at a product company; trusted skills: TensorFlow, Recommendation Systems, NLP; 30-day notice, open to work, based in Trivandrum.
- #2 Senior Machine Learning Engineer (7.2 yrs); career history shows building retrieval/search & ranking/recommendation systems in production with NDCG/MAP-style evaluation; at a product company; trusted skills: scikit-learn, Recommendation Systems, QLoRA; 15-day notice, open to work, based in Noida.
- #3 Lead AI Engineer with 6.7 yrs; career history shows building retrieval/search & ranking/recommendation systems in production with NDCG/MAP-style evaluation; at a product company; trusted skills: Vector Search, Recommendation Systems, Qdrant; 30-day notice, open to work, based in Jaipur.
- #4 Senior NLP Engineer, 8.9 yrs; career history shows building retrieval/search & ranking/recommendation systems in production with NDCG/MAP-style evaluation; at a product company; trusted skills: Pinecone, PEFT, Elasticsearch; 30-day notice, open to work, based in Coimbatore.



REPRODUCIBLE & DEFENSIBLE

Built for Stage-3 reproduction and the interview

- **One command, CPU-only, no network at ranking time.**
python rank.py --candidates candidates.jsonl --out submission.csv (~3 min, < 5-min budget)
- **Every score is inspectable.**
Interpretable sub-scores per candidate — easy to defend in the architecture interview.
- **No LLM at inference → scales to a 200K pool in production.**
Exactly the latency/quality thinking the JD asks a Senior AI Engineer to demonstrate.
- **Honest, varied, fact-grounded reasoning passes manual review.**
Specific facts + JD connection + acknowledged concerns + rank-consistent tone.